

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering

Course Code & Title: CS8451 & Design and Analysis of Algorithms

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Innovative Practice: Flipped Classroom

Topic: Maximum Matching in Bipartite Graph

Type of Learning:

Collaborative Learning

Learning Objectives:

O1: To make the students to understand the concept of maximum matching in Bipartite graph.

Description:

A flipped classroom is a type of collaborative learning where students might watch videos took the notes according to their understanding level at home and then do the task provided with questions and at least some background knowledge in the classroom.

Uses of Flipped Classroom:

- It helps students to allow self-paced learning.
- It increases individual attention.
- It helps the students to be better prepared on the particular topic.

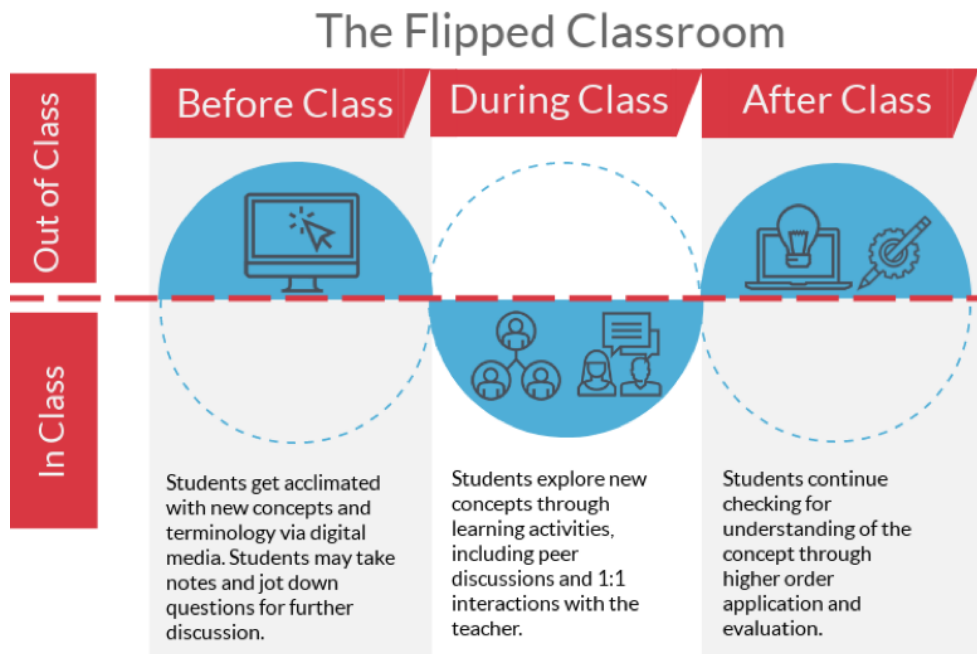
Justification for chosen the topic:

Maximum matching in bipartite graphs is one of the important topics, repeatedly asked in university questions. This activity makes the students to get a sound knowledge in this concept. Students can prepare individually about a topic by watching the lecture video and share their ideas with their classmates that enhance their knowledge and oral communication skills.

Implementation of Flipped Classroom:

- Learning materials such as video and documents were sent to the student's mail and Canvas, one week before conducting this activity.

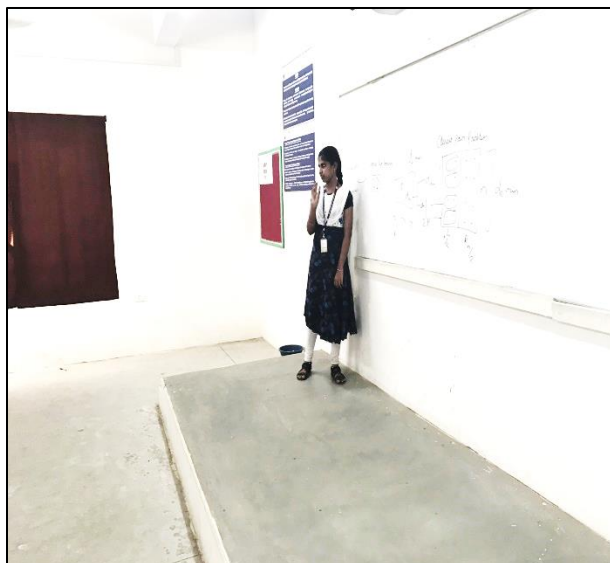
- Instruct the students to watch the video individually at their home and take the notes according to their understanding level.
- On the day of the activity, small teams were formed by their own interest.
- Each student in a group discusses among the peer members based on their self-learning and write the points for Presentation within 10 minutes.
- Once they prepared the content for presentation with their peer members, each group present the given topic for maximum of 5 minutes.



(Credit: Google Image)

Glimpse of Flipped Class Activity:





Outcomes:

- This activity actually increases attentiveness in the class and also improves the students' skill of critical analysis of the topics.
- This activity makes the students to understand the topic clearly.
- The students could be able to enrich various skills such as communication, coordination among team members.

Relevance to POs

Objective	PO1	PO2	PO3	PO4	PO5	PO9	PO10
O1	3	3	2	2	2	3	2

Reflective Report:

Identified Problems

- Some team members are hesitated and lack to share the concepts that they learnt and discussed.

Initiatives to address the problems

- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.
- Addressed the students that the activity is not a kind of evaluation methodology to assess.

Post-implementation

- Students were actively participated in this activity.

- From this activity, the students can get more clarity in the particular topic by discussing and sharing their views with the other students in the class.
- Learning at home automatically becomes student-centric rather than teacher-centric.

References:

- https://www.ritrjpm.ac.in/images/computer-science/41_IT6801_FlippedClassroom.pdf
- <https://www.ritrjpm.ac.in/images/computer-science/CS8603-FlippedClassroom.pdf>

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